

Weekly Report
Peter-Emil Georgiev
Prof. Rao
Senior Project

This week, I am getting started on choosing the exact algorithms that I will be implementing for my senior project. I have stopped at three:

- implementing a catmull-rom spline for color adjustment ,
- gaussian blur
- seam carving — this will be more complicated to make.

More about the algorithms, which links with information about them is at the bottom of the report.

I also wrote a first draft of the Introduction section of the thesis, which I will be attaching additionally to this file.

Additionally, I will be implementing version control and for that I will also need to implement a graph data structurer that will store all changes made. As such, I would also like to implement a merging algorithm, that would allow the merging of multiple unrelated image versions.

For next week, I am going to be working on streamlining my ideas for the project further and starting to work on the app itself — I would like to have a app which can import images , as I want to be able to test the algorithms as I implement them.

Information about the Algorithms:

Catmull-rom spline:

While general color adjustments, i.e contrast, highlights and shadow adjustments are generally pretty simple to create, as they are mostly just linear transformations of the matrix, which is the image, which is why I will be making it a bit more complicated through adding an adjustment curve. This will be done using a Catmull-rom spline, which, from what I gather is generally what is used. Additionally I can use that for resizing images as well, which would be a 2D extension of the algorithm.

What is a Catmull-rom spline? It is a curve interpolation algorithm, which creates a continuous curve from a few ‘checkpoints’, which in my case would be the adjustment nodes (i.e. exposure, highlights and shadows). In resizing, this follows the same principle, but the ‘nodes’ would be the old pixels, and the smooth curve would be the new pixels that need to be added in order to make the resizing look smooth.

<https://www.cs.cmu.edu/~fp/courses/graphics/asst5/catmullRom.pdf>

2. Gaussian blur:

I will also be implementing a Gaussian blur brush. The math for it is listed here:
<https://www.w3.org/Talks/2012/0125-HTML-Tehran/Gaussian.xhtml>

I will also be utilizing a similar algorithm to use for light normalization across the image.

3. Seam carving algorithm for image resizing

Identifying the least important pixels (seams) and removing them for the purpose of image resizing. This will be more complicated to make efficiently.

<https://www.geeksforgeeks.org/python/resize-an-image-using-seam-carving-algorithm-and-streamlit/>

Github repo: